TRAMONTANA ICS-01
MONITORING REPORT - MP2

Project title	Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India
Project ID	4457
Monitoring period	24-August-2023 to 12-September-2024
Original date of issue	01-August-2025
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Version	1.3
VCS Standard Version	VCS Standard v4.7
Prepared by	Tramontana Asset Management

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Over the second monitoring period (MP2) between 24 August 2023 and 12 September 2024, a further 69,441 ICS were installed in households, taking the total number of ICS distributed under the project to 115,601. During the period, distribution also began in the second project Instance area located in the Rampachodavaram division of Alluri Sitarama Raju district of Andhra Pradesh state. 53,480 of the 69,441 ICS installed in the monitoring period were distributed in this new area. The other 15,961 ICS were installed in the first project Instance area. Information on ICS installation and each beneficiary is recorded in the distribution database provided (see "ICS-01 Distribution Database M2").

Table 1. Planned distribution schedule of ICS across the project area corresponding to instances under the grouped project

Instance	Planned ICS Distribution
Region 1	50,000
Region 2	100,000
Region 3	100,000
Region 4	100,000
Region 5	50,000

Table 2. Ex-post emissions reduction table for the first and second monitoring periods

Monitoring period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Reduction VCUs (tCO_2e)	Total VCUs (tCO_2e)
MP1 21-January-2023 to 23-August-2023	52,560	21,894	1,533	29,133	29,133
MP2 24-August-2023 to 12-September-2024	311,990	130,890	9,055	172,045	172,045
Total	364,550	152,783	10,588	201,178	201,178

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Joint Validation + Verification	21-January-2023 – 23-August-2023	VCS	Earthood Services Private Limited	0.586
Verification	24-August-2023 – 12-September-2024	VCS	Carbon Check (India) Private Limited	1.058

1.3 Sectoral Scope and Project Type

Sectoral scope[1]	03 – Energy Demand
Project activity type	Project Type: II – Energy Efficiency Improvement Project Grouped Project: Yes

1.4 Project Proponent

Organization name	Tramontana Impact Limited (subsidiary of Tramontana Asset Management)
Contact person	Dr Ramanan M. Loganathan
Title	Director
Address	2nd Floor, Gaspé House, 66-72 Esplanade, St. Helier, Jersey, JE1 1GH
Telephone	+44 (0)203 773 0160
Email	impact@tramontana.co.uk

1.5 Other Entities Involved in the Project

Organization name	Tramontana Sustainable Commodities Limited (subsidiary of Tramontana Asset Management and formerly called CropZone Agro Forestry Private Limited)
Role in the project	Local Implementation

[1] Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes

Contact person	Mr Saibaba Abburi
Title	Director
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Email	saibaba@cropzone.in

1.6 Project Start Date

Project start date	21-January-2023
Justification	Date of installation of the first ICS under the grouped project

1.7 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	21-January-2023 to 20-January-2030

1.8 Project Location

All project activity under the grouped project is within the geographical boundaries of India, as depicted in the Project Description Document (PDD), with approximate coordinates of latitude 21.555808 and longitude 78.746972.

The first and second project instances are within the Alluri Sitarama Raju (ASR) district of Andhra Pradesh state shown in Figure 1[2]. Specifically, Instance 1 is located within the Paderu division (approximate coordinates of latitude 18.011959 and longitude 82.555030) and Instance 2 within the Rampachodavaram division (approximate coordinates 17.4500, 81.7667). Both divisions are shown in Figure 2. The elevation of these regions is approximately 1000 metres above sea level.

[2] https://www.india.com/news/india/andhra-pradesh-to-have-13-new-districts-from-april-4-check-list-of-total-26-districts-here-5317197

Figure 1. Map of Andhra Pradesh state in India divided by district where the first and second project activity instances are located (see blue arrow)

Figure 2. Project activity area split by division and instance (Paderu – Instance 1, Rampachodavaram – Instance 2)

1.9 Title and Reference of Methodology

Type	Reference ID	Title	Version
Methodology	VM0050	Energy efficiency and fuel switch measures in cookstoves	1.0
Methodology	CDM AMS-II G	Energy efficiency measures in thermal applications of non-renewable biomass	13.1

VCS Tool	VT0008	Additionality Assessment	1.0
CDM Tool	CDM TOOL30	Calculation of the fraction of non-renewable biomass	4.0

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.10.2 Registration in Other GHG Programs

Was the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.12 Sustainable Development Contributions

During the monitoring period, activities contributing to the Sustainable Development Goals (SDG) were conducted across two areas of the Alluri Sitarama Raju (ASR) district of Andhra Pradesh state, as described in section 1.8. Across both areas, the project contributes to environmental, social, economic and technological benefits which lead to the sustainable development of the local area and country as follows:

Environmental Benefits

- Preservation of forest stock and protection of ecosystems, habitats and soil health by reducing the rate of deforestation and forest degradation across the project area.
- Reduction in toxic indoor air pollution from the cleaner combustion characteristics and increased thermal efficiency of the project ICS. This lowers health issues from respiratory and other diseases. Communities also save on medical expenditure and require less time off to look after unwell family members.
- Lower GHG emissions generated by a reduction in the use of non-renewable biomass.
- Less water and effort needed for cleaning utensils and maintenance of the household as a result of the cleaner cooking process and minimised smoke/soot generation.

Social Benefits

- More hours for skill/economic development and children learning owing to reduced time spent sourcing firewood.
- Improvement to gender equality as it is primarily women that carry out the tasks relating to stove usage, firewood collection and cleaning.
- Improved home conditions and quality of life for families by reducing localised indoor air pollution and from the time savings.
- Enhanced safety and reduced burns/injuries for children owing to the compact nature of the Tramontana ICS and less direct exposure to the flame.
- Confidence and comfort as no material change is required to traditional cooking practices.
- Development of IT skills for the local teams employed as a result of training and experience gained under the project activity.

Economic Benefits

- Employment opportunities and skill development for the local communities and economy through the project activity. This includes specifically employing and training local teams for operational tasks such as distribution, monitoring, maintenance, data analysis and conducting stakeholder meetings.
- More hours available for productive income generating activities and children learning as a result of reduced household work and fuel sourcing.

Technological Benefits

- Introduction of cleaner technology directly to rural communities.

- Employment of local workers and provision of training and experience in the use of tablets with an Android software application to conduct baseline surveys, distribution and monitoring work.
- Provision of training and experience in IT for local office teams to undertake the tasks around project data compilation and analysis.

Nine of seventeen global UN SDGs have been identified as relevant to this project and are illustrated with black boxes below. Table 3 provides the detailed description of each of these SDGs including the relevant baseline scenario, corresponding contribution from the project activity during the second monitoring period and overall contributions across the project lifetime.

Table 3: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions (Monitoring Period 2)	Contributions over project lifetime (Monitoring Periods 1 and 2)
1)	1.4	Project Specific Indicator: Number of people provided access to the basic service of cleaner cooking through the provision of a Tramontana ICS	Implemented activities to increase	100% of beneficiaries included in this monitoring period had a Below Poverty (BPL)/ration card issued by the state government. These cards are only given to the most marginalised and underrepresented communities in India. Each card has been photographed and recorded for every beneficiary in the distribution database. It was observed from the baseline surveys for Instance 2 that the: • average annual income of project users was INR 38,614 or equivalently USD1.23 per day, • 70% of households were using traditional fixed mud clay stoves and 30% using three-stone stoves, highlighting the poverty levels in the region. In relation to community status, • 100% of beneficiaries in this monitoring period confirmed belonging to the indigenous Scheduled Tribe/Adivasi or Scheduled Caste/Dalit communities. Through the provision of ICS to the communities, direct access to the basic service of cleaner cooking has been provided. Over this monitoring period a total	Across the project lifetime, 100% of beneficiaries have been recorded to have a BPL card. All cards have been photographed and recorded in the distribution database. In relation to community status, all beneficiaries apart from three (who belong to the Other Backward Class community) confirmed belonging to the indigenous Scheduled Tribe/Adivasi or Scheduled Caste/Dalit communities. To date, through the project a total of 406,136 people have been provided with direct access to cleaner technology.

				406,136 people now have direct access to cleaner energy/technology through the project.	
2)	3.9	Project Specific Indicator: Proportion of households reporting reduced volumes of smoke inside their homes in addition to general health improvements	Implemented activities to decrease	Upon using an ICS, of the monitored households in the second monitoring period: • 100% stated that the local smoke pollution in their homes had decreased • 100% confirmed experiencing general health improvements In addition, the portable nature of the ICS offers the flexibility to beneficiaries to use their ICS outside of their house. These benefits have resulted in communities experiencing less of the physical and mental health issues which were stated in the baseline database. During monitoring surveys 99.7% of beneficiaries stated that the function of their ICS was 'excellent' or 'good'. The remaining 0.3% (3 beneficiaries) reported damage to their ICS. Although the stoves were still being used, the ICS were repaired in line with the high-standards of the Project Proponent (see section 3.1 for details). In the comments section of the database, statements have been made by users on the improvements experienced to well-being and health, including reduced respiratory and retinal issues.	Across the households across both monitoring periods: • 100% stated that the local smoke pollution in their homes had decreased • 100% confirmed experiencing general health improvements Across all monitoring periods, all except three beneficiaries stated that the function of their ICS was 'excellent' or 'good'. These three beneficiaries were from this second monitoring period and as discussed here reported damage to their ICS (see section 3.1 for details).

3)	4.3	Project Specific Indicator: Proportion of households reported to be spending more time on education as a result of time savings generated by the project	Implemented activities to increase	Improving the access to education in the local communities is a useful aspect of the project, particularly given the lack of access and opportunities available to the community members and their children. From the baseline database it was observed that: • 30% of children are currently involved in stove related tasks, thereby preventing them from focussing on education. Monitoring for the period highlighted that with the time saved by using an ICS: • 45.3% of respondents indicated that their family (primarily children) were already able to spend more time in education. Throughout the ICS distribution phase and during all post-distribution engagements, information and support is provided on a continuous basis to all beneficiary households on the correct use of the ICS with proper fuel preparation, as well as educating them on the importance of employing sustainable practices. During monitoring, many beneficiaries also commented (see monitoring database) that they were	Across the project life • 53.8% of respondents indicated that their family (primarily children) were already able to spend more time in education.

				aware of the local area project coordinators if ICS support was needed.	
4)	4.4	Project Specific Indicator: Number of adults given specific ICT training under the project	Implemented activities to increase	All 319 employees under the project (155 new in this monitoring period, see SDG 8) were given extensive training for their roles, involving use of tablets with the Tramontana application and methods to ensure high data quality/integrity. Those working on computers were given additional IT and data analysis training.	A total of 319 employees have been given technical training, including use of the Tramontana app, under the project.
5)	5.4	Project Specific Indicator: Proportion of beneficiaries able to spend more time contributing to the main family income, spending time with family and helping others in the community	Implemented activities to increase	Gender equality is a critical aspect of the project given the disproportionate number of female members who perform stove related tasks. It was observed from the baseline database that in: • 97.3% of households the adult female family member is the primary stove user • 91.0% of households women are relied upon to collect the firewood Records highlighted that family members, mainly women, were now able to: • spend more time contributing to the main family income source in 98.1% of cases • spend more time with the family in 88.1% of cases • in 16.5% of cases commence community work.	Family members across the project have now been able to: • spend more time contributing to the main family income source in 98.8% of cases • spend more time with the family in 91.5% of cases • in 14.2% of cases commence community work. Across all the teams employed, 13% are female. The Project Proponent is

				Photos of female family members using their ICS, such as those presented in Figure 4, are recorded in the monitoring database.	seeking to hire and train more female members where possible as the project expands.
6)	7.1	7.1.2 Proportion of the population with primary reliance on clean fuels and technology	Implemented activities to increase	It was observed that • 100% of the households were currently using their ICS and had been using it over the monitored period. Upon summing all family members across households that have received an ICS in the database, and factoring in an adoption factor for those that are using their ICS, it is calculated that cumulatively 406,136 people (243,593 additional this Monitoring Period) have direct access to cleaner energy/technology now through the project. This represents approximately 40% of the Alluri Sitharama Raju population.	Across the project, a total of 406,136 people have been provided with direct access to cleaner technology to date. This represents approximately 40% of the Alluri Sitharama Raju population.
7)	8.5	Project Specific Indicator: Number of people employed directly and indirectly as a result of the project	Implemented activities to increase	The employment and training of people in country has resulted in skill development and improvements to the livelihoods of local people. This in turn contributes to local and national economic growth, as well as development of the work force. During this monitoring period, the project has directly employed: • 85 people full-time • 242 people part-time	The project has in total directly employed: • 85 people full-time • 242 people part-time. 22 full-time employees have been given senior roles as mandal coordinators. Many more people have been indirectly employed through

				The majority of these employees are from the local project region. Employees have gained experience conducting baseline, distribution and monitoring surveys for the project which is recorded in the databases. In total 55 tablets were provided to the local teams to carry out the field surveys. Of the full-time employees, 11 were given senior roles as Mandal Coordinators to take charge of the field team in each district of the second project Instance region. Mandal Coordinators employed from the local communities also themselves use an ICS and have first-hand experience of the benefits to their family's lives. The number of indirect employees (manufacturing and logistics) has had little change during this monitoring period.	the project activity. These include: • 150 people employed in the country for manufacturing of the project ICS • 68 employed for transporting the ICS across the country.
8)	12.2	Project Specific Indicator: Total mass of firewood saved as a result of the project activity	Implemented activities to increase	Baseline studies for Instance 2 found that: • 100% of families source firewood for their stoves from local forest areas in the project region. Based on the average firewood consumption used by traditional stoves reported in the baseline database of 4.83 tonnes per year for Instance 1, 6.04 tonnes per year for Instance 2, and using the methodology (15% baseline efficiency) and technical specification of the Tramontana ICS detailed in section 1.12 of the PDD,	Across baseline studies for Instances 1 and 2, 100% of families source firewood from local forest areas and a total of 224,393 tonnes of firewood has been saved under the project (see calculations in sheet "ICS-01 Fuel Savings M2").

				we calculate a total saving of 189,806 tonnes of firewood during the monitoring period (see sheet “ICS-01 Fuel Savings M2” provided).	
9)	13.0	13.2.2 Total greenhouse gas emissions per year	Implemented activities to decrease	As presented in section 1.1, the • emissions reductions for the monitoring period is 172,045 tCO_2e. All ICS distributed are currently being used by families on a regular basis. • No additional monitoring was required for this SDG as the calculation is already a central requirement to verification under the Verra standard.	The total emissions reductions from all operational ICS is 201,178 tCO_2e.
10)	15.1	Project Specific Indicator: Forest area conserved due to reduced firewood consumption from project implementation	Implemented activities to increase	As highlighted for SDG12, baseline surveys confirm that all firewood sourced for use in stoves is collected from local forest areas and contributes to widespread deforestation in the region. From baseline surveys it is observed that • the average distance to collect firewood is 3.15km with 96% of respondents in Instance 2 stating the distance is increasing year-on-year. Starting from the monitoring output for SDG12, the total mass of firewood saved over the period was calculated as 189,806 tonnes. Assuming average dimensions for a forest tree in the local region, a total of 2,623 hectares of forest area has been estimated	Across baseline studies for Instances 1 and 2: • The average distance to collect firewood has been observed as 3.26km. The total mass of firewood saved is 224,393 tonnes, which amounts to an estimated 3,101 hectares of forest being preserved (see calculations in sheet “ICS-01 Fuel Savings M2”).

				to be preserved over the period through implementing the project (see the sheet "ICS-01 Fuel Savings M2" provided).	

1.13 Commercially Sensitive Information

All information for the purposes of project description and project implementation is included in this report which is made publicly available.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholders have been identified for the second project instance, adopting a similar approach to that for the first instance but incorporating region specific attributes.

Stakeholder Identification	Pre-baseline surveys were first conducted across the second instance region to identify all eligible households. Subsequently multiple beneficiary stakeholder meetings were held to further engage with potential eligible ICS users and convey more detailed information. Other project stakeholders are directly employed or partnered with at the design and planning phase of a project instance. <u>Primary stakeholders</u>: local families receiving a Tramontana ICS, local employees such as field staff, local administrators (e.g. District Collectors), NGOs <u>Secondary stakeholders</u>: Outside of the local project areas, additional people are employed to support the project activities including manufacturing, transportation etc.
Legal or customary tenure/access rights	As part of the project activities, close engagement with the local communities is facilitated to identify any potential concerns and legal conflicts in relation to the project. By means of access to a local project representative or through a designated phone number, stakeholders can raise any concerns or questions they have throughout the project. To date, no conflicts or issues have been raised. The project does not interfere with any legal or customary rights to territories and resources held by project stakeholders.
Stakeholder diversity and changes over time	The ICS beneficiaries under the project live in rural areas and are from indigenous and socio-economically disadvantaged communities. The vast majority are represented by the Scheduled Tribe/Adivasi or Scheduled Caste/Dalit community categories. The

	makeup of the indigenous people within a project instance is similar and all follow similar customs and practices. This has not changed since verification. As part of the project, local staff from the project region are employed and trained to undertake the activities on-site and engage with the communities. All primary and secondary stakeholders involved in the project are located in India.
Expected changes in well-being	The project aims to have a positive impact on stakeholder wellbeing. The social and economic benefits of the project are outlined in detail in Section 1.12 and are briefly listed here: • Reduced drudgery and exposure to risks including wild animal attacks, injuries from tree cutting/wood chopping • More hours available for productive income generating activities and schooling • Improved home conditions and quality of life for families • Enhanced safety and reduced burns/injuries • Development of IT skills for the local teams employed • Employment opportunities
Location of stakeholders	ICS beneficiaries under the second project instance are located in the Rampachodavaram division of ASR district in Andhra Pradesh state in India. The precise location of each beneficiary is recorded during ICS installation and is presented in the distribution database. Outside of the local project areas, the people employed to support the project activities (e.g. manufacturing, transport) are all located within the project boundary of India.
Location of resources	Many of the project communities undertake subsistence farming on land which has been provided to them by the government. This land is typically in close proximity to the beneficiary's home. The project does not interfere with such territories of the stakeholders.

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	155 local people (51 full-time, 104 part-time) have been employed during this monitoring period to ensure the continued use of the ICS by beneficiaries and for regular engagement with the communities. All concerns or questions that occur following stakeholder meetings can be directly reported to a local mandal representative or voiced through a dedicated phone number. These formats are suitable given many local people in the communities are illiterate and unable to express their feedback in written format. All points raised will be added by the team to a local grievance register. The information in the register will be reviewed on a regular basis by the Project Proponent, with all concerns addressed and incorporated into the project on a merit basis. Additionally, at least annual visits are being made by the local teams to the villages to monitor the performance of each ICS. Every visit is recorded via the Tramontana Application and collated into a larger Internal Monitoring Database, allowing the Project Proponent to continuously review and improve the overall project performance.
Date(s) of stakeholder consultation	The stakeholder consultations for the second Instance comprising of 11 mandals in Rampachodavaram division of ASR district in Andhra Pradesh state occurred between the 8th November 2023 and 28th November 2023. One meeting was conducted per mandal ensuring a uniform and extensive consultation with community stakeholders. Across the second Instance area, a total of 4550 local community members attended the meetings. Minutes and attendance sheets were taken for all meetings and are provided to the VVB. A summary of the schedule and attendance for all stakeholder meetings is also outlined in Appendix 1.
Communication of monitored results	The monitored results have been communicated to all relevant stakeholders through local coordinators. For example, monitoring surveys highlighted that a small number of beneficiaries had been leaving their ICS outdoors during the times when not in use. This resulted in some rust forming on the ICS after rainy periods. Consequently local coordinators reminded beneficiaries to always bring their ICS inside following usage.

Consultation records	A full list of all attendees at each meeting is recorded, including their signatures, contact number and address. In addition, the minutes of every meeting including stakeholder questions and the responses are recorded and documented (translated into English). All information recorded from meetings including detailed minutes, attendance sheets, stakeholder questions and photos are made available.
Stakeholder input	All questions raised and corresponding answers given by the project team during stakeholder meetings are recorded in minutes. All minutes and attendance sheets for the second project instance are provided to the VVB and Verra. Some commonly asked questions from the second project instance are provided in Table 4. The project team also made themselves available after all meetings for any further discussion points to ensure all attendees were satisfied. No changes were required to be made to the project design as a result of the queries/discussions. Any key points raised from future stakeholder meetings will be recorded and if justified incorporated into the project design.

Figure 3: Stakeholder meetings for Instance 2 depicting a Tramontana presentation, special guest speeches, high turnout, ICS demonstrations and community Q & A sessions

Table 4: Questions raised by local community members and responses given by the project team for various stakeholder meetings under the second project Instance

Name of attendee	Question	Local coordinator response
T. Ammajii from Barrimaamidi village (Gangavaram)	How many times can this improved cookstove be used in a day?	The Tramontana cookstove is designed for daily use. It is made from high-quality materials such as steel/ceramic, which improve the durability, efficiency and

		performance. It is designed to handle different types of cooking tasks throughout the day, making it reliable for everyday use
M. Chinnalaxmi from Devara Madugula village (Addateegala)	Will the improved cookstove resolve the issue of excessive smoke produced by our three-stone stove during cooking?	Yes, the cookstove addresses this issue and significantly reduces smoke during cooking as result of its improved design
N. Rajulamma from Kindra village (Rajavommangi)	Can these cookstoves be employed both inside the house and outdoors?	Yes, these stoves are portable, allowing for use both indoors and outdoors
D. Kumari from Maredumilli village (Maredumilli)	Is there ongoing support or follow-up after receiving the cookstove?	Yes, we provide ongoing support to ensure a continued, optimal performance of the cookstove. This may include periodic check-ins, free-of-cost maintenance, and opportunities for additional training or resources as needed
S. Ramulamma from Kothagudem village (Kunavaram)	How do these stoves benefit us?	As mentioned during the meeting and on the banners around the venue, these stoves offer: a cleaner environment, healthier indoor air, time savings in fuel collection/stove use, potential cost-efficiency and overall better community well-being. The time savings for example, can enable you to earn more income from doing more agricultural work on your farm.

2.1.3 Free, Prior, and Informed Consent

Consent	During stakeholder meetings, the benefits of the ICS devices are described and safe usage of the ICS is demonstrated. The meetings also represent an opportunity for stakeholders to raise any concerns or ask questions, as discussed in Section 2.1.2. These measures ensure that stakeholders are well informed, able to relay their views, and project activities are fully transparent. During distribution of the ICS, upon voluntary agreement to participation, an agreement is signed between the Project Proponent and each beneficiary. The terms of the agreement are outlined verbally in detail during ICS distribution.
Outcome of FPIC	All beneficiaries provided consent to their involvement in the project and the terms of the license agreement. The feedback was overwhelmingly positive for project participation based on the stakeholder meetings and pre-baseline surveys conducted. The project has not encroached on land, relocated people without consent, or forced physical or economic displacement. No conflicts have been recorded under the project to date.

2.1.4 Grievance Redress Procedure

Grievances received	Resolution and outcome
4	From the sample over the monitoring period, four stoves were reported damaged. Following the process outlined in section 3.1, the stoves were promptly repaired by the local team using stored spare parts. Information including the ICS QR code, date of repair, parts replaced, and the field team member responsible for conducting the service have been recorded in the grievance register. Records of these repairs are also accessible online through the Project Proponent's maintenance database.

2.1.5 Public Comments

Summary of comments received	Actions taken
NA	No comments were received as part of the stakeholder consultation or outside of the public comment period

2.2 Risks to Stakeholders and the Environment

During this monitoring period, a new project area, Instance 2, was introduced, bringing additional stakeholders into the project. Although this group have been newly engaged, they share many similarities with stakeholders in Instance 1 in terms of demographic profile, economic conditions, and everyday cooking challenges. Additionally, the project technology distributed in the region, the CRS-23 stove, remains unchanged from Instance 1.

As with Instance 1, in Instance 2 we have specifically hired people from the local community to ensure optimal engagement and support. These employees have be trained under the same programme proven successful previously, and are managed by the Project Proponent. A field office has also been set up in the new region. The existing grievance redress process has been extended to Instance 2, and engagement activities such as stakeholder meetings and access to field staff have been carried out in line with the robust protocol and standards previously implemented. This continuity ensures that communities in Instance 2 receive the same level of support as those in Instance 1.

The consistency in stove technology, project approach and community profile help ensure that existing safeguard structures address the needs of both areas. As a result we can apply section 2.2 of the PDD as also relevant for this monitoring period.

2.3 Respect for Human Rights and Equity

The following section has not changed during the monitoring period. Please see section 2.3 of the PDD for details.

2.4 Ecosystem Health

The following section has not changed during the monitoring period. Please see section 2.4 of the PDD for details.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

Up to the 12th September 2024, a total of 115,601 ICS have been distributed in total under the project.

Specifically during the second monitoring period (MP2), an additional 15,961 ICS were distributed in the Instance 1 region (taking the total there to 62,121 ICS), whilst from 7th January 2024 distribution began in the Instance 2 region. So far a total of 53,480 ICS have been distributed under MP2 in the Instance 2 region.

Distribution of each stove was carried out by local field staff and corresponding data recorded using the Tramontana ICS software application as described in the PDD. All 155 new employees in Instance 2 have been trained under an extensive programme, involving: engagement techniques with the community, approaches to minimising errors, and methods to deal with issues such as non-response cases, conflicts of interest and interviewer bias.

Figure 4: Beneficiaries photographed with their Tramontana ICS during the second monitoring period

During our monitoring assessments, 4 stoves were found to be damaged (QR codes: A1A0009680, A1A0032246, A1A0042033 and A1A0058493). These stoves were repaired on the 25th September, 20th September, 7th November and 11th October 2024 respectively and details are provided in "ICS-01 Monitoring Database M2". It was confirmed by the individual beneficiaries, that at no point were the damaged stoves not being used and as a result GHG emissions reduction calculations are not affected.

Generally, there were no significant events which occurred during the monitoring period which affected the GHG emission reductions or required a change to the monitoring framework presented. There has also been no change to the entities involved in the project including the Project Proponent.

3.2 Deviations

3.2.1 Methodology Deviations

The project activity has not taken any deviations from the applied Verra methodology VM0050.

3.2.2 Project Description Deviations

No Project Description deviations have been applied during this monitoring period.

3.3 Grouped Projects

For the second monitoring period, 53,480 stoves have been distributed in Instance 2. The new Instance meets the eligibility criteria of VCS Standard v.4.7[3] as presented in the following tables.

Eligibility criteria	Compliance by Instance 2
Meet the applicability conditions set out in the methodology applied to the project	Instance 2 under the grouped project meets the conditions of the methodology as specified in section 3.2 of the PDD
Use the technologies or measures specified in the project description	Instance 2 employs the technology type where an energy efficient ICS replaces the traditional inefficient cookstoves used by households. Information on the ICS technology used for all instances is provided in the PDD.
Apply the technologies or measures in the same	Only thermally efficient ICS is distributed to households in Instance 2, and no other technologies are adopted

[3] https://verra.org/wp-content/uploads/2024/04/VCS-Standard-v4.7-FINAL-4.15.24.pdf

manner as specified in the project description	
Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area	A separate baseline campaign has been conducted for Instance 2 to provide a more accurate representation for the region. The baseline scenario for the Instance is discussed in section 3.4.
Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area	Instance 2 has additionality characteristics consistent with Instance 1, and is demonstrated using the activity methodology below. Step 1: Regulatory Surplus Instance 2 project work is not mandated by any law, government policy or regulation in the host country Step 2: Positive List Instance 2 complies with the positive list under the methodology (see section 3.5 of the PDD for details) Step 3: Project Method Not applicable

In addition to the criteria above, Instance 2 meets the specific criteria required for any new instances (added after validation) as presented below.

Eligibility criteria	Compliance by new project instances
Occur within one of the designated geographic areas specified in the project description	Instance 2 is located within the Rampachodavaram division of the ASR district of Andhra Pradesh state, which is within the geographical boundary of India (see section 1.8).
Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial	Instance 2 complies with the full set of eligibility criteria for inclusion under the existing project activity as described in the table above.

conformance with multiple sets of eligibility criteria is insufficient	
Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate conformance with the applicable set of eligibility criteria and enable evidence gathering by the validation/verification body	Extensive information related to Instance 2 has been provided in this monitoring report. In addition: minutes from all stakeholder meetings, the instance specific baseline and distribution databases, detailed methodology calculations are all provided and submitted to the VVB.
Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions)	A standardised license agreement is signed physically (with written signature or thumbprint) voluntary by each ICS beneficiary. It includes (amongst other conditions) provisions for ownership of all emissions reductions generated by the ICS to be solely with the Project Proponent. The agreement is generated through the Tramontana ICS software application on a tablet on which it is signed by the household end user. Signed pdf copies of all agreements are made available in the distribution database.
Have a start date that is the same as or later than the grouped project start date	Instance 2 distribution began on the 7th January 2024, which is after the start date of the grouped project of 21st January 2023.
Only be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project, through to the	The Instance 2 start date is 7th January 2024. As this is after the start of this verification period (24th August 2023), crediting is to begin from 7th January 2024.

end of the total project crediting period	
Not be or have been enrolled in another VCS project	Instance 2 has not and will not be enrolled in any other VCS project
Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 – 3.6.9 of the VCS Standard v4.7	In line with CDM AMS-II G, a single project activity shall not exceed the equivalent of 60 GWh per year or 180 GWh thermal per year in fuel input. We calculate that the energy savings for a single project activity within Instance 2 does not exceed 60 GWh per year. The calculations meeting such condition are given in the Emissions Reductions spreadsheet provided.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☒ Yes ☐ No

The baseline for Instance 1 was assessed during the first monitoring period. Over the second monitoring period, ICS distribution commenced in a second project instance area. Despite both project instances sharing comparable characteristics, a separate baseline survey has been conducted for Instance 2 in order to gauge an accurate representation of the current practices in the specific region. The selection and justification of the baseline scenario is unchanged from the PDD and the detailed approach which was used for Instance 1, including the data collection and analysis methods, have also been applied here. Note the baseline for Instance 1 remains unchanged and still relevant to the project, so no changes to the PDD or baseline emissions for Instance 1 were required.

Following initial pre-baseline assessments conducted between August and October 2023, baseline surveys for the second project instance were conducted across 11 mandals in the Rampachodavaram division of Alluri Sitarama Raju district in Andhra Pradesh state. These were conducted prior to distribution between 12th December 2023 and 6th January 2024. Applying random sampling to the pre-baseline data, 100 villages were randomly selected for baseline assessment and within each village 10-15 surveys were conducted. This resulted in a total of 1472 baseline surveys being completed, significantly exceeding the minimum requirement specified in the PDD (see section 4.3.1). The use of significant oversampling has generally been applied across the project to maximise the reliability of the data.

From baseline survey results for Instance 2 (see database provided), it was found that the majority of families (70%) were using traditional fixed mud clay stoves and a smaller proportion (30%) using three-stone stoves (Figure 5). These findings are supported by the 2019-2021 India National Family Health Survey[4], which found that, of those using solid fuels, 99.3% are using open fire or mud-clay stoves for cooking.

It was observed that 100% of surveyed households used their stove for: cooking, preparing tea/coffee and water boiling, whilst 79% also used it for heating purposes. The traditional stoves were found to be used an average of 3.7 hours per day all year round and 10.3 hours on average per week was spent sourcing, carrying and processing (e.g. cutting, shredding) the firewood. Over 83% of survey respondents were using their stove inside the household, but only 34% had a provision for smoke to escape (e.g. opening in walls or a window).

In all cases firewood was reported to be sourced directly from forests. Although some government reports suggest there is high LPG usage in Andhra Pradesh, schemes such as PMUY, have been particularly unsuccessful in remote areas, such as in Instance 2, for the same reasons described in section 3.4 of the PDD.

Following the release of the VM0050 methodology in October 2024, 154 Kitchen Performance Tests (KPTs) and respective baseline surveys were conducted from 22nd November 2024 to 16th December 2024 and 17th to 22nd June 2025. Note that the additional baseline surveys were conducted to include two further questions required by the new methodology:

1. How many meals did you prepare yesterday? Is this typical?
2. Is there any difference with the amount of meals prepared during a typical weekend day?

From the KPTs conducted, it was measured that an average of 16.55kg was used per household per day in the Instance 2 area. This equates to 6.04 tonnes per household per year. Note that this is 20.3% greater than in Instance 1 and thereby results in a 25.9% higher baseline emissions in the area.

In line with the *Correction and Clarifications to VM0050*[5] document, the measured baseline consumption is to be compared to data from reputable sources including “government publications, peer-reviewed literature, third-party studies, or official data and statistics” which must have been collected “no more than five years ago”. While national-level data has been updated through various reports and surveys, state-level breakdowns post-2013 are scarce. The India State of Forest Report and National Sample Survey Office (NSSO) are the most reliable sources, but their latest publicly available datasets on fuelwood consumption remain based on earlier assessments. Consequently, as presented in section 5.3.2 the PDD, an independent set of KPTs were conducted in-situ by a specialist external company Koshish Sustainable Solutions, with vast experience of biomass cookstoves testing both lab and in-situ. Their assessment found

[4] https://www.dhsprogram.com/pubs/pdf/FR375/FR375.pdf

[5] https://verra.org/wp-content/uploads/2025/02/CC_VM0050_v1.0_Feb2025.pdf

an average of 14.5kg of wood per household per day being used in the baseline scenario. Although these tests were conducted in Instance 1, the project Instances share similar characteristics in terms of: region (same district - ASR), demographics and stove usage behaviours. The differences from the 14.5kg measured by Koshish and 16.55kg measured during the KPT can be explained primarily due to the accessibility of wood. The forest cover in Instance 2 is much denser than in Instance 1, making firewood more readily accessible. This often leads to higher consumption due to the availability at no cost. Additionally, some forest protections rules in Instance 1 have prevented households from felling trees in certain areas. These protection rules are not as strict in Instance 2. The combination of firewood being more abundant and fewer restrictions are likely the reason for higher usage in Instance 2.

Of the 154 households in Instance 2 where KPTs were conducted, 59 remain outside of the project (i.e. not receiving a Tramontana ICS). These households, known as “control households”, are used to monitor any changes in stove usage characteristics in the Instance 2 project area. If any changes are observed over time, for example, reduced fuel consumption or the adoption of an additional stove, the baseline fuel consumption for Instance 2 ($BC_{b,i,y}$) will be updated.

Of the KPT households surveyed, the average household size (Hh_i) was found to be 3.13 equivalent standard male adults, equivalent to 4 people. The 2020-2021 Multiple Indicator Survey in India[6] reported that the average household size in rural areas of Andhra Pradesh is 3.5, thereby supporting our observations. The parameter also meets the 90/10 precision/confidence requirement of the methodology as described in section 4.3.1.

Every user carried their firewood by headload and the average distance from the source location was 3.15km. 96% of respondents confirmed the distance to the source location has been increasing year-on-year, as forest areas closer to households have been deforested. It is exclusively women who were using the stoves (aside from a few cases where no women lived in the household), with the firewood gathering also undertaken by them, sometimes with the help of other family members.

A number of key issues were widely reported across baseline surveys. These included: regularly experiencing wild animal attacks in the forest, difficulties in finding sufficiently large volumes of wood, toxic localised smoke causing breathing issues, burning eyes and dizziness, and physical pain in undertaking the physically demanding tasks of cutting trees and carrying firewood long distances in hilly areas.

Almost all respondents (98%) stated that there is a lot of localised smoke produced by their traditional stoves and that the associated tasks were time consuming, giving little time for other activities. Despite all respondents knowing that kitchen smoke has a negative impact on health, they have little ability to control it. When questioned on why they use traditional stoves which have such issues, 99.8% stated that they were not aware of any other stove types, 99.7% that

[6] https://mospi.gov.in/sites/default/files/publication_reports/MultipleIndicatorSurveyinIndiaf.pdf

firewood is abundantly available and 76.9% that their current stove has been at least manageable in terms of handling/maintenance.

All respondents confirmed having a Below Poverty Line/ration card, which has been photographed at the time of surveying and is recorded in the database. Over 92% of the respondents stated that their family's main work is in agriculture and the total income per family was an average of USD1.23 per day.

Over 99% of respondents stated that they were not happy with their current stove and accordingly very few stated that it was not a priority to change. All participants confirmed that they did not own an ICS and would be interested to receive a new ICS if offered one.

Figure 5: Photos from the baseline database depicting a traditional mud clay stove, three stone fire, and local community members cooking with significant smoke pollution being generated

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

The parameters in this sub-section have not changed from those described in section 5.1 of the PDD, however for completeness it has been included here.

Data / Parameter	$EF_{b,i,CO2}$
Data unit	tCO_2/TJ
Description	CO_2 emission factor for fuel used by baseline device type i in the baseline scenario
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas[7]
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the baseline scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of baseline emissions
Comments	--

Data / Parameter	$EF_{p,j,CO2}$
Data unit	tCO_2/TJ
Description	CO_2 emission factor for fuel used by project device type j in the project scenario
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas[7]
Value applied	112
Justification of choice of data or description of	Wood is the fuel used in the project scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied

[7] https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/2_Volume2/19R_V2_2_Ch02_Stationary_Combustion.pdf

measurement methods and procedures applied	
Purpose of data	Calculation of project emissions
Comments	--

Data / Parameter	$EF_{b,i,nonCO2}$
Data unit	tCO_2e/TJ
Description	Non-CO_2 emission factor for fuel used by baseline device type i in the baseline scenario
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas[7]
Value applied	9.46
Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the baseline scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of baseline emissions
Comments	--

Data / Parameter	$EF_{p,j,nonCO2}$
Data unit	tCO_2e/TJ
Description	Non-CO_2 emission factor for fuel used by project device type j in the project scenario
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas[7]
Value applied	9.46
Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the project scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of project emissions

Comments	--

Data / Parameter	$NCV_{b,i}$
Data unit	TJ/tonne
Description	Net Calorific Value of baseline fuel used by baseline device type i
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas[7]
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the baseline scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of baseline emissions
Comments	--

Data / Parameter	$NCV_{p,j}$
Data unit	TJ/tonne
Description	Net Calorific Value of project fuel used by project device type j
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas[7]
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the project scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of project emissions
Comments	--

Data / Parameter	$BC_{ex-ante,b,i}$
Data unit	tonnes
Description	Ex-ante average quantity of fuel used per baseline device type i
Source of data	Option 1 as specified in the VM0050 v1.0 methodology guidelines: Kitchen Performance Test (See "ICS-01 KPT Instance 1")
Value applied	4.83
Justification of choice of data or description of measurement methods and procedures applied	The parameter value used is from the measured in-situ KPT results for 151 households sampled from the first project instance. The baseline KPT for Instance 1 was conducted between 13th November and 13th December 2024 and 11th to 16th June 2025. KPTs were exclusively conducted by local employees who received specialist training in line with the Kitchen Performance Protocol v4.0 published in March 2018. Based on the sampling calculation outlined in Appendix 5 of the methodology, a minimum sample size of 36 with 90/10 confidence precision was calculated (see section 5.3.2 of the PDD for detailed calculations). KPTs were conducted for 151 households in Instance 1, therefore achieving an oversampling rate of 4.2. Detailed analysis has been conducted on the parameter which is presented in the excel sheet "ICS-01 KPT Instance 1" provided and in section 5.3.2 of the PDD. We observed a standard deviation 0.9894, confidence interval 0.265 and precision (2-sided) 3.35%. Therefore the 90/10 requirement of the methodology is achieved.
Purpose of data	Calculation of baseline emissions
Comments	Note that 51 of the KPTs conducted are in households which remain outside of the project (i.e. not receiving a Tramontana ICS) and act as "control households" in accordance with the methodology.

Data / Parameter	$\eta_{old,avg}$
Data unit	Fraction

Description	Weighted average efficiency of baseline devices that are replaced by project devices
Source of data	Option 1 as specified in the VM0050 v1.0 methodology
Value applied	15%
Justification of choice of data or description of measurement methods and procedures applied	The default value for three-stone fires using firewood or cookstoves with no improved combustion air supply or flue gas ventilation has been selected
Purpose of data	Calculation of baseline emissions
Comments	--

4.2 Data and Parameters Monitored

The parameters in this sub-section have been updated for the second monitoring period from those described in section 5.2 of the PDD.

Data / Parameter	$N_{j,k,y}$
Data unit	Number
Description	Number of commissioned project devices of type j from batch k in year y
Source of data	Distribution database (See "ICS-01 Distribution Database M2")
Description of measurement methods and procedures to be applied	Distribution records are created through surveys conducted when each ICS is successfully distributed to a project beneficiary at their household This parameter is not calculated on a sample basis, but instead on the total project population. Thus it does not form part of sample-based monitoring in section 4.3.
Frequency of monitoring/recording	At the time of commissioning/distribution of ICS
Value monitored	115,601
Monitoring equipment	Distribution records are created using Samsung Galaxy A9+ tablets with the Tramontana ICS software application. Submitted records are stored locally on the tablet and automatically synced to a cloud database once the device is

	connected to the internet in the local offices. The automation of syncing maintains the data integrity and minimises any risk of data loss, theft, tampering or errors in translation. Further details on data reliability and traceability are explained in section 5.3.2 of the PDD
QA/QC procedures to be applied	During distribution an extensive dataset is obtained on each end user, including personal details, ID card/community certificate information and photo, and a photo of the individual together with their new ICS
Purpose of the data	Calculation of baseline and project emissions for the monitoring period
Calculation method	--
Comments	--

Data / Parameter	$n_{j,k,y}$
Data unit	Fraction
Description	Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y
Source of data	Monitoring (See "ICS-01 Monitoring Database M2")
Description of measurement methods and procedures to be applied	The parameter is monitored based on representative sampling of the distribution records in the database. As per the methodology, the sampling complies with the latest version of CDM guidelines[8]. Through the surveys, beneficiaries are questioned on their ICS usage practices over the monitoring period. If it is found that a non-project stove is being used following conducting monitoring surveys, the parameter $n_{j,k,y}$ is adjusted using a ratio of usage frequency of the project ICS compared to the frequency with which all stoves are used over a defined time period. For instance for a given household, if it is found from a survey that a non-project stove is being used 7 times per week and the ICS is being used 21 times a week, the following factor would be used as part of calculating $n_{j,k,y}$:

[8] https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

	$$q = \frac{21}{21+7} = 0.75$$ The values across all households monitored is then averaged to obtain a value for $q_{j,k,y}$
Frequency of monitoring/recording	At least annually
Value monitored	The value monitored during the period was 1.0 as all ICS samples monitored were operational owing to regular maintenance/repair and full replacement if necessary. However, in line with the cap described in the *Correction and Clarifications to VM0050*[5] document published in February 2025, a value of 0.9 is applied. The mean and standard deviation of the data collected is calculated as 1 and 0 respectively.
Monitoring equipment	Trained field staff inspect the household to identify whether the ICS is in operation including taking photos of the stove, its components and the cooking area. A detailed survey using the Tramontana ICS software application is also conducted with the end user to identify current usage patterns and operation/performance over the past year.
QA/QC procedures to be applied	Access to the relevant experienced local staff member and their phone number is provided to every end user in case of any problems faced. This ensures that the Project Proponent can be directly contacted in order to resolve any issues promptly and make sure stoves remain in operation. Before conducting any monitoring surveys, field staff are also given specific training on the process and purpose of visits. This facilitates standardisation and a high quality being maintained. In addition to the required mandatory monitoring under the methodology, local staff complete additional internal checks with the aim of covering the entire population annually. Note that in this process, the sampled monitoring households are also revisited to verify the accuracy of existing data collected. More generally, the project aims to exceed the required sample confidence thresholds by using techniques such as oversampling.
Purpose of the data	Calculation of baseline and project emissions
Calculation method	Based on sample-based monitoring, the parameter is calculated as:

	$$n_{j,k,y} = q_{j,k,y} \times \frac{n_{op,j,k,y}}{n_{tot,j,k,y}}$$ where n_{op} = number of operational ICS monitored n_{tot} = total number of ICS monitored
Comments	--

Data / Parameter	Hh_i $Hh_{j,k}$
Data unit	Equivalent standard male adults
Description	Average household size of the target population using baseline device type i Average household size of the target population using project device type j from batch k
Source of data	Baseline and Project KPT surveys (See "ICS-01 KPT Instance 2")
Description of measurement methods and procedures to be applied	During KPT surveys, the household composition of families is recorded. The equivalent standard male adult per household is then identified, as per *Guidelines for Woodfuel Surveys for FAO* referenced in the methodology, and an average taken to find the average household size.
Frequency of monitoring/recording	New values are determined every time the project activity is undertaken in a new instance
Value applied	For the first project instance: $Hh_i = 3.38$ (includes control households) $Hh_{j,k} = 3.47$ For the second project instance: $Hh_i = 3.13$ (includes control households) $Hh_{j,k} = 3.09$
Monitoring equipment	Surveys using tablets with the Tramontana ICS software application

QA/QC procedures to be applied	The value obtained has been compared to the average household size in the region as reported by the latest official government census. The project aims to exceed the required sample confidence threshold through using techniques such as oversampling.
Purpose of data	Estimation of average energy consumption per household, and for cross-checking of energy and fuel consumption values
Calculation method	For each family, the household composition of the family is recorded, converted into the equivalent male adult units as per the Kitchen Performance Test Protocol v4.0, and then used in the calculation of the average across all KPT households
Comments	The values are instance-specific and will be measured for each individual instance

Data / Parameter	n_{HH}
Data unit	Number
Description	Number of ICS distributed to each household
Source of data	Distribution database (See “ICS-01 Distribution Database M2”)
Description of measurement methods and procedures to be applied	When an ICS is distributed by a field team member, the QR code is scanned and a corresponding family record is created and synced into the distribution database. Each family record is associated with a unique BPL card (see section 5.3.1). When there are multiple ICS records linked to a single BPL card, it signifies that multiple ICS have been distributed to a single family. Generally, if a family has six or more members, they may request a second stove This parameter is not calculated on a sample basis, but instead on the total project population. Thus it does not form part of sample-based monitoring in section 4.3.2.
Frequency of monitoring/recording	At the time of commissioning/distribution of ICS
Value monitored	1 as standard. If a family requests more than one ICS, the Project Proponent will consider providing an additional ICS if, amongst other additional details obtained, the amount of

	firewood used is measured to be significantly higher than the average across the mandal.
Monitoring equipment	Surveys using tablets with the Tramontana ICS software application
QA/QC procedures to be applied	--
Purpose of the data	Calculation of baseline and project emissions for the monitoring period
Calculation method	--
Comments	--

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass that is established to be non-renewable used by device in year y
Source of data	Option 2 as specified in the VM0050 v1.0 methodology
Description of measurement methods and procedures to be applied	Calculated by determining the share of non-renewable woody biomass in the total quantity of woody biomass consumed for each project instance. The parameter value is obtained by using CDM Methodological Tool 30[9] with inputs specific to the local state of the project instance. An uncertainty discount of 26% is then applied to the calculated value
Frequency of monitoring/recording	Determined ex-ante for a project Instance and fixed for the crediting period
Value monitored	A value of 0.510 was calculated for the Instance 1 area. Please refer to section 4.4 of the PDD and the spreadsheet "ICS-01 fNRB Calculation Instance 1". For Instance 2, a value of 0.556 is calculated – please refer to the spreadsheet "ICS-01 fNRB Calculation Instance 2"
Monitoring equipment	The value for average consumption of wood fuel per household has come from the KPT tests described in section 5.3 of the PDD. All measurements have been taken by extensively trained

[9] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf

	field teams who follow KPT Protocol v4.0 and use digital scales to directly record the weight of wood. Moisture is also accounted for through the use of digital moisture meters. Other inputs to the parameter calculation have been obtained from the IPCC Guidelines for National Greenhouse Gas Inventories, the 2019 India State of Forest Report, the 2019-21 Andhra Pradesh National Family Health Survey and 2011 Census of India Population Projection. Note that all references and calculations are described in detail in the spreadsheets "ICS-01 fNRB Calculation Instance 1" and "ICS-01 fNRB Calculation Instance 2".
QA/QC procedures to be applied	To ensure the reliability of NRB and RB inputs, data has been crosschecked with other sources and publications available. These crosschecks include research from university studies as well as government publications. These are also presented in respective spreadsheets "ICS-01 fNRB Calculation Instance 1" and "ICS-01 fNRB Calculation Instance 2"
Purpose of the data	Calculation of baseline and project emissions
Calculation method	The value as per CDM Methodological Tool 30: Calculation of the fraction of non-renewable biomass[18] is calculated using: $f_{NRB} = \frac{NRB}{NRB + RB} \times (1 - 0.26)$ where NRB is the quantity of non-renewable biomass consumed in tonnes and RB is the quantity of renewable biomass in tonnes in the area. The factor 26% is applied under the latest methodology.
Comments	--

Data / Parameter	$BC_{b,i,y}$
Data unit	Tonnes
Description	Average quantity of fuel used per baseline device type i during year y
Source of data	Option 1 as specified in the VM0050 v1.0 methodology (See "ICS-01 KPT Instance 2")
Description of measurement methods and procedures to be applied	The parameter is measured by conducting in-person KPTs in sample households for each project Instance. Each test lasts 4 days and, in line with the methodology, the result is scaled using the average household size.

	For each project Instance, a minimum of 50 households will remain outside of the project as "control households". If, throughout the project lifetime, it is observed that fuels, fuel sources or technologies used by these households undergo any changes, the baseline consumption value used for emission reductions calculations will be updated. The parameter is measured separately for each instance to gauge an accurate representation of fuel use in the specific region.
Frequency of monitoring/recording	Every five years
Value monitored	For Instance 1, an average value of 13.22kg per day was measured per baseline stove, equivalent to 4.83 tonnes annually. In Instance 2, an average value of 16.55kg per day was measured per baseline stove, which is equivalent to 6.04 tonnes annually.
Monitoring equipment	Field teams directly record the weight of the wood using digital scales. Moisture is measured using a digital moisture meter. The digital scales used by the field team are the ATOM A 302 Electronic Digital Hanging Stainless Steel Hook Luggage Portable Scale with resolution 0.01kg and accuracy +/-0.01kg The moisture meters used are the Signette-MT10-R-1 with resolution 0.1% and accuracy +/-0.5%. All measurements are recorded using the Tramontana ICS application and stored in the cloud database.
QA/QC procedures to be applied	The digital moisture meters and weighing scales have been calibrated by the manufacturer to help ensure accuracy of measurements taken. Field staff conducting the tests are given KPT specific training ensuring the accuracy and reliability of the results. The test results from household batch must meet the 90/10 confidence/precision as set by the methodology, else more sample tests are undertaken until the requirement is met. This is discussed further in section 5.3 of the PDD.
Purpose of the data	Calculation of baseline emissions for the monitoring period
Calculation method	--

Comments	The parameter is dynamic and is chosen to be measured separately for each instance in order to obtain an accurate representation of fuel use in the specific region.

Data / Parameter	$\eta_{new,avg,y}$
Data unit	Fraction
Description	Weighted average efficiency of project devices in year y
Source of data	Option 1 as specified in the VM0050 v1.0 methodology (See Appendix 2)
Description of measurement methods and procedures to be applied	The parameter is measured by conducting Water Boiling Tests (WBT) involving averaging the cold-start and hot-start high-power phases of the ICS. In line with the methodology, for an ICS of type i and vintage y a sample from the first ICS batch of 3 ICS are selected and for each one 3 WBTs are conducted. The degradation of efficiency measured from monitoring the first ICS batch may be applied across all batches. The WBT is conducted by an independent technical laboratory and certified test results will be provided as part of monitoring.
Frequency of monitoring/recording	Annually
Value monitored	For y=1, the thermal efficiency of the CRS-23 project measured from tests conducted on 6 randomly selected operational project ICS resulted in an average value of 34.73% Note that up to the end of this monitoring period, only the CRS-23 model has been distributed to households.
Monitoring equipment	Laboratory tests
QA/QC procedures to be applied	Experienced technical laboratories following ISO standards are selected for the purposes of conducting all WBTs. The test results from the samples must meet the standard deviations limit and 90/10 confidence/precision as set by the methodology, else more sample tests will be undertaken until the requirement is met. This is discussed further in section 4.3 of this report.
Purpose of the data	Calculation of baseline and project emissions for the monitoring period

Calculation method	For an ICS of type i and vintage y, the parameter is determined by averaging the results from independent WBTs conducted on a minimum sample of 3 operational ICS
Comments	--

Data / Parameter	$BC_{p,j,k,y}$
Data unit	Tonnes
Description	Average quantity of fuel used by project device type j from batch k during year y
Source of data	Option 1 as specified in the VM0050 v1.0 methodology (See "ICS-01 KPT Instance 2")
Description of measurement methods and procedures to be applied	The parameter is measured by conducting in-person KPTs in sample households participating in the project activity. Each test lasts 4 days and, in line with the methodology, the result is scaled using the average household size. For this monitoring period, a total of 294 KPTs have been conducted across the project Instances. Specifically, 195 project KPTs were conducted as paired samples (with baseline KPTs), and 99 KPTs were conducted on ICS randomly selected from the database "ICS-01 Monitoring Database MP2".
Frequency of monitoring/recording	At least biennially
Value monitored	For Instance 1, an average value of 4.55 kg per day was obtained for the project ICS, which is equivalent to 1.66 tonnes annually Whilst for Instance 2, an average value of 4.98 kg per day was obtained for the project ICS, which is equivalent to 1.82 tonnes annually
Monitoring equipment	Field teams directly record the weight of firewood using digital scales. Moisture is recorded using a digital moisture meter. The digital scales used by the field team are the ATOM A 302 Electronic Digital Hanging Stainless Steel Hook Luggage Portable Scale with resolution 0.01kg and accuracy +/-0.01kg The moisture meters used are the Signette-MT10-R-1 with resolution 0.1% and accuracy +/-0.5%.

	All measurements are recorded using the Tramontana ICS application and stored in the cloud database.
QA/QC procedures to be applied	The digital moisture meters and weighing scales have been calibrated by the manufacturer to help ensure accuracy of measurements taken. Similar to baseline KPTs, field staff conducting the tests are given KPT specific training ensuring the accuracy and reliability of the results. The test results must meet the 90/10 confidence/precision as set by the methodology, else more sample tests will be undertaken until the requirement is met. This is discussed further in section 5.3 of the PDD. Results are also compared to published literature/government publications before being submitted.
Purpose of the data	Calculation of baseline and project emissions for the monitoring period
Calculation method	--
Data / Parameter	--

4.3 Monitoring Plan

The sampling plan detailed in section 5.3 of the PDD has been applied for this monitoring period.

4.3.1 Sample design/sizing and Field data

The parameters monitored over this monitoring period are outlined in Table 5.

Table 5. Parameters monitored during the monitoring period

Parameter	Description
$n_{j,k,y}$	Proportion of commissioned project devices of type j in batch k operating during year y
$BC_{b,i,y}$	Average quantity of fuel used per baseline device type i during year y
$BC_{p,j,k,y}$	Average quantity of fuel used by project device type j from batch k during year y

Hh_i / $Hh_{j,k}$	Average household size of the target population using baseline device i / project device j from batch k
$\eta_{new,avg,y}$	Weighted average efficiency of project devices in year y

Field monitoring surveys were conducted to determine the first parameter in Table 5, $n_{j,k,y}$, in line with section 5.3 of the PDD. Over the monitoring period, 1188 surveys were completed and $n_{j,k,y}$ was calculated as 1 (see ICS-01 Monitoring Database M2). However, in line with the *Correction and Clarifications to VM0050*[5] document published in February 2025 the parameter must be capped at 0.9 upon ensuring the following actions are undertaken and demonstrated:

1. *Selection of technologies and fuels that fully meet the cooking needs of the target population, demonstrated by citing robust research or conducting an investigation of cooking practices and attitudes*
2. *Implementation of support activities to assist the target population in effectively operating and maintaining their cookstove. These may include providing and maintaining their cookstoves. These may include providing materials (print, in-person, or video) on how to operate the cookstove to prepare common local foods, how to troubleshoot common operational issues, and how to make minor repairs (including obtaining necessary replacement parts). All such communications and materials must be provided in local language(s) commonly used in the project area*
3. *Provision of a commonly used, toll-free communications channel through which the target population can contact the project proponent to access support (e.g. maintenance and repair services)*

As outlined in the PDD, the project has been designed to meet and exceed requirements related to technology selection, user support, and accessibility. Starting with the first condition above, Instance 2 was introduced into the project during the monitoring period following a thorough assessment of local cooking practices, fuel use, and household energy needs. The findings confirmed that the existing project cookstove model met the cooking requirements of the new Instance area, including for common meal preparation and water boiling.

To satisfy criteria 2, project implementation continues using the Project Proponent's established approach. Proven initiatives from Instance 1 have been replicated in Instance 2 to ensure ICS sustained use and success. A new field office and a dedicated repairs team have been set up to deliver maintenance services at the same high standard and responsiveness seen in Instance 1. Households also benefit from consistent support in operating and maintaining their cookstoves, with team members hired locally to provide support. Instructional materials covering proper usage and troubleshooting have been distributed through printed handouts and in-person sessions, all contributing to the sustained use of the ICS and overall project success.

To satisfy criteria 3, a toll-free communications channel operates across both Instance areas, allowing community members to contact project staff for guidance, maintenance, and grievance redress. Engagement activities such as stakeholder meetings and access to local staff remain aligned with the protocols and standards that were established in Instance 1. All three criteria stated above are met and therefore we adopt the capped usage rate for $n_{j,k,y}$ equal to 0.9.

The next two parameters in Table 5, $BC_{b,i,y}$ and $BC_{p,j,k,y}$, are monitored using the approach described in Kitchen Performance Test Protocol v4.0[10] and methodology VM0050 v1.0. Specifically a paired-sample study has been conducted, first for households in the baseline scenario and then in the post-distribution project scenario. Each of the KPTs lasted four days and were mostly conducted Monday-Thursday with weekends typically avoided. Festivals and other holidays where there may be above-average fuel consumption were also avoided.

To select the specific households in Instance 2 for surveying, a simple random sampling approach was followed, in accordance with section 5.3.2 in the PDD. Specifically, each household from the baseline survey was assigned a unique identifier. Using Microsoft Excel's RANDARRAY function, 95 random numbers were generated within the range of the list size, ensuring no duplicates. These values were then matched against the corresponding individuals on the list to form the final sample. This approach guaranteed that each household had an equal chance of being selected, resulting in a representative and unbiased sample.

In addition to the paired samples discussed above, 59 baseline KPTs were conducted in households in Instance 2 which will remain outside of the project (i.e. not receiving a Tramontana ICS). These households act as the “control households” for Instance 2. If any changes in these control households are observed, for example, change in stove practices or fuel use, the baseline fuel consumption ($BC_{b,i,y}$) for the instance will be updated to reflect this change.

As previously presented in the PDD and in spreadsheet “ICS-01 KPT Instance 1”, the KPT results for Instance 1 gave $BC_{b,i,y} = 4.83$ tonnes/year. During the second monitoring period, distribution began in Instance 2 and therefore baseline KPTs were conducted in this region. From a total of 154 baseline KPTs conducted (including control households), $BC_{b,i,y}$ was measured as 16.55 kg/day or equivalently 6.04 tonnes/year (see “ICS-01 KPT Instance 2” sheet).

In line with the methodology, the test results from the above households are required to meet a minimum 90/10 confidence/precision. From the baseline KPT results, the mean and standard deviation used to calculate $BC_{b,i,y}$ is calculated from the 154 samples (n) as 5.50 kg/pp/day and 1.78 kg/pp/day respectively. From these, the confidence interval (CI) and precision are calculated as follows:

$$CI = mean \pm 1.645 \times \frac{SD}{\sqrt{n}}$$

Note that 1.645 represents the 90% confidence required by the VM0050 v1.0 methodology.

The upper and lower bound of CI are calculated and the difference is outputted:

$$CI = 5.50 \pm 1.645 \times \frac{1.78}{\sqrt{154}}$$

[10] https://cleancooking.org/binary-data/DOCUMENT/file/000/000/604-1.pdf

$$\Delta CI = 0.471$$

$$Precision = 0.5 \times \frac{\Delta CI}{mean}$$

$$= 0.5 \times \frac{0.471}{5.50}$$

$$= 4.28\%$$

As the precision is less than 10%, the mandatory 90/10 rule has been met.

For this monitoring period, project KPTs were conducted in Instance 1 and Instance 2 to calculate the values of $BC_{p,j,k,y}$ for each area respectively. The KPT results for Instance 1 gave $BC_{p,j,k,y} =$ 1.66 tonnes/year (see "ICS-01 KPT Instance 1" sheet) and for Instance 2 gave $BC_{p,j,k,y} = 1.82$ tonnes/year (see "ICS-01 KPT Instance 2" sheet).

In line with the methodology, the project KPT test results are required to meet a minimum 90/10 confidence/precision. From the 150 samples (n) in Instance 1, the mean and standard deviation was measured to be 1.34 kg/pp/day and 0.34 kg/pp/day. The CI and precision are calculated as follows:

$$CI = 1.34 \pm 1.645 \times \frac{0.34}{\sqrt{150}}$$

$$\Delta CI = 0.092$$

$$Precision = 0.5 \times \frac{0.092}{1.34}$$

$$= 3.41\%$$

As the precision is less than 10%, the mandatory 90/10 rule has been met.

From the 144 project KPTs (n) conducted in Instance 2, the mean and standard deviation is calculated as 1.64 kg/pp/day and 0.45 kg/pp/day respectively. The CI and precision are then determined as follows:

$$CI = 1.64 \pm 1.645 \times \frac{0.45}{\sqrt{144}}$$

$$\Delta CI = 0.122$$

$$Precision = 0.5 \times \frac{0.122}{1.64}$$

$$= 3.72\%$$

As the precision is less than 10%, the mandatory 90/10 rule has been met.

These calculations are outlined in detail in the spreadsheets "ICS-01 KPT Instance 1" and "ICS-01 KPT Instance 2" along with the supporting data.

The fourth parameter set in Table 5 Hh_i /$Hh_{j,k}$ was also monitored during the KPT surveys. Specifically the household composition of each family was recorded, converted into the equivalent male adult units as per the Kitchen Performance Test Protocol v4.0, and finally averaged across all KPT households. To first ensure the baseline parameter Hh_i (and baseline survey outputs) for Instance 2 meets the 90/10 confidence/precision requirements from the methodology and CDM guidelines, the mean and standard deviation are calculated from the 154 samples (n) discussed above, as 3.13 and 1.14 respectively. Following the same approach used for $BC_{b,i,y}$ and $BC_{p,j,k,y}$, the CI and precision are calculated as follows:

$$CI = 3.13 \pm 1.645 \times \frac{1.14}{\sqrt{154}}$$

$$\Delta CI = 0.302$$

$$Precision = 0.5 \times \frac{0.302}{3.13}$$

$$= 4.83\%$$

Given the precision output is less than 10%, the <u>90/10 rule has been met,</u> and therefore the parameter Hh_i (and baseline survey outputs) for Instance 2 meet the requirements of the methodology and CDM guidelines.

Similarly for $Hh_{j,k}$, representing the average household size of the population using the project device, the mean and standard deviation calculated from the 150 project KPT samples for Instance 1 (n) were 3.47 and 1.23 respectively. The CI and precision are therefore calculated as:

$$CI = 3.47 \pm 1.645 \times \frac{1.23}{\sqrt{150}}$$

$$\Delta CI = 0.331$$

$$Precision = 0.5 \times \frac{0.331}{3.47}$$

$$= 4.77\%$$

As the precision is less than 10%, again the <u>90/10 rule has been met.</u>

Finally, for Instance 2 the mean and standard deviation calculated from the 144 project KPT samples (n) were 3.09 and 1.11 respectively. The CI and precision are therefore calculated as:

$$CI = 3.09 \pm 1.645 \times \frac{1.11}{\sqrt{144}}$$

$$\Delta CI = 0.305$$

$$Precision = 0.5 \times \frac{0.305}{3.09}$$

$$= 4.94\%$$

As the precision is less than 10%, again the 90/10 rule has been met.

The final parameter in Table 5 $\eta_{new,avg,y}$ was monitored using a separate approach as per the methodology. The methodology required efficiency tests to be conducted for those stoves which have been used for (at least) one year (y=1). WBTs were undertaken by a specialist independent technical laboratory Koshish Sustainable Solutions Pvt Ltd during September 2024. The average for y=1 was determined from 18 samples (6 ICS with 3 WBTs conducted for each), which were selected based on the Simple Random Sampling approach described in section 5.3.2 of the PDD. The QR codes and efficiencies for the 6 ICS are listed below in Table 6. All test results (including corresponding QR codes) are presented in Appendix 2.

Table 6. List of ICS selected for WBTs and their efficiency results

Test Number	ICS QR Code	WBT Efficiency Result
1	A1A0024142	34.7%
2	A1A0031378	34.7%
3	A1A0008699	34.6%
4	A1A0007441	34.9%
5	A1A0019697	35.0%
6	A1A0010594	34.5%
Average		34.73%

In line with CDM Standard for Sampling and Surveys[8], the Student's t-distribution must be adopted for sample sizes less than 30 to ensure the 90/10 confidence/precision level is met. From the 18 samples (n) in Appendix 2, the mean efficiency and standard deviation (SD) are calculated as 34.73% and 0.23 respectively. Using these parameters and assuming a 90% confidence level, the critical t-value (T) is found to be 1.74[11]. The margin of error (ME) can therefore be calculated as:

$$ME = T \times \frac{SD}{\sqrt{n}}$$

$$ME = 1.74 \times \frac{0.23}{\sqrt{18}}$$

$$= 0.094$$

[11] https://www.ttable.org/

ME can then be used in combination with the mean to find the CI and precision of the data collected, as follows:

$$CI = mean \pm ME$$

Note that upper and lower bounds of CI are calculated here with the difference outputted:

$$CI = 34.73 \pm 0.094$$

$$\Delta CI = 0.187$$

$$Precision = 0.5 \times \frac{\Delta CI}{mean}$$

$$= 0.5 \times \frac{0.187}{34.73}$$

$$= 0.27\%$$

As the precision is less than the 10% threshold, the mandatory 90/10 rule has been met.

4.3.2 Sample size

Specific to this monitoring period, the minimum sample size required to calculate $n_{j,k,y}$ for the period must be determined.

Simple Random Sampling was used to determine the sample size for the parameter $n_{j,k,y}$ in Table 5. The following equation from Annex 5 of the VM0050 v1.0 methodology was used:

$$n \geq \frac{1.645^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 \times p(1-p)} \quad \text{Equation 1}$$

Where:

n	=	Sample size for households
N	=	Total number of households
p	=	Expected proportion of parameter of interest
1.645	=	Represents the 90% confidence required
0.1	=	Represents the 10% relative precision

In line with the methodology, a 90% confidence level is selected, with a value of $p = 0.75$ applied for parameter $n_{j,k,y}$. The parameter p represents the proportion of traditional (or other) stoves that are *not* being used. This value is very conservative given any ICS observed to have issues has been continuously repaired/replaced and all traditional cookstoves were demolished/removed.

Applying Equation 1 with a population size 115,601 for the monitoring period, we calculate the minimum sample size required for $n_{j,k,y}$ for a 90/10 confidence level/precision as:

$$n \geq \frac{1.645^2 \times 115{,}601 \times 0.75(1-0.75)}{0.1^2 \times 0.75^2 \times (115{,}601-1) + 1.645^2 \times 0.75(1-0.75)}$$

$$n \geq 90.13$$

$$n = 91$$

Over the monitoring period, 1188 surveys have been conducted as presented in the database. This represents a factor 13 times the required sample size.

To ensure the parameter $n_{j,k,y}$ calculated from the 1188 monitoring surveys also meets the 90/10 confidence/precision level required by the methodology, the mean and standard deviation are calculated. The mean was calculated as 1.0 and standard deviation 0 as all ICS samples monitored were found operational owing to regular maintenance and repairs being conducted. As the standard deviation is 0, the result meets the 90/10 confidence/precision level required by the methodology.

4.3.3 Implementation plan & Data quality control

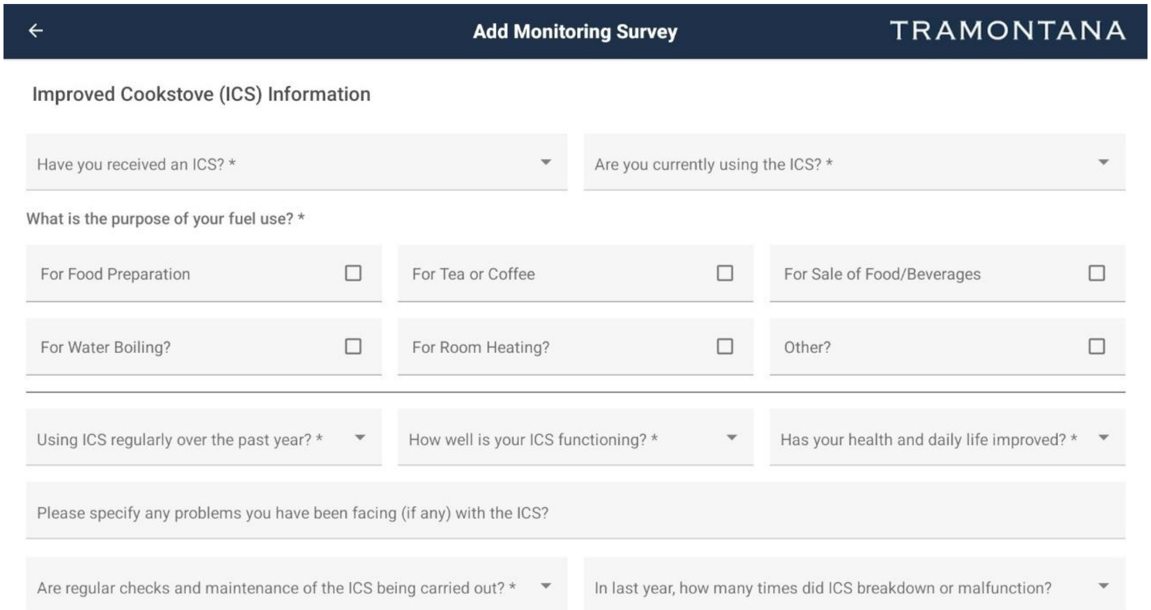
We follow the general processes described in section 5.3.2 of the PDD relating to implementation and data acquisition over this monitoring period.

Site visits were undertaken to visually examine households and conduct surveys (including to assess the first parameter in Table 5). The final monitoring data is recorded in the database provided and includes (i) information on the ICS installed over the period and (ii) monitoring data records from field surveys. The data is extensive and directly attributable to each individual ICS/household, helping form clear evidence for any emissions reductions that are claimed over the period.

All field staff that were employed during the monitoring period have been trained under an extensive programme prior to commencing any surveys. Surveys were conducted using the Tramontana ICS software application (Figure 6) and all entries were rigorously checked by the office team for errors or inconsistencies. Any errors or inconsistencies found were immediately resurveyed.

As presented in the monitoring database for this period, only 4 of the monitored stoves (0.34%) were reported to be damaged. Each of these stoves was promptly repaired in line with the maintenance protocol described in section 4.3, and it was confirmed that at no point were the damaged stoves not being used. Additionally there was no evidence of any use of alternative stoves which may, for instance, have been built or purchased since receiving a project ICS. This is reflected in the outputs in section 4.2.

Further to our official monitoring efforts, the Tramontana team undertake additional extensive internal monitoring to cover the full project population. Each household is aimed to be visited at least once a year by the local field team, and information regarding ICS use, performance, and any feedback is collected. During these visits, visual checks are carried out to ensure beneficiaries are using their ICS, and to check for evidence of any other stoves used. To date, there has been no evidence of other stoves being used, and 100% of internally monitored households are using their ICS (including after any repairs) and confirm they will continue to do so in the future.



Add Monitoring Survey
TRAMONTANA

Improved Cookstove (ICS) Information

Have you received an ICS? *
Are you currently using the ICS? *

What is the purpose of your fuel use? *

For Food Preparation
For Tea or Coffee
For Sale of Food/Beverages
For Water Boiling?
For Room Heating?
Other?

Using ICS regularly over the past year? *
How well is your ICS functioning? *
Has your health and daily life improved? *

Please specify any problems you have been facing (if any) with the ICS?

Are regular checks and maintenance of the ICS being carried out? *
In last year, how many times did ICS breakdown or malfunction?

Figure 6: Screenshot from the Tramontana application used by field staff to record all user data for the purposes of monitoring

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

The following section describes how emissions reductions are calculated for the project over the second monitoring period (24th August 2023 – 12th September 2024). To calculate the emission reductions, first the baseline, project and leakage emissions are determined separately in sections 5.1 – 5.3 below. Once identified, the overall emission reductions are calculated and presented in section 5.4.

5.1 Baseline Emissions

To calculate the baseline emissions for the second monitoring period, the approach described in section 4.2 of the PDD, using Equation 2, is followed in addition to the specific calculations described below.

$$\boldsymbol{BE_y = \sum_{i,j,k} EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO2} \times f_{NRB,y} + EF_{b,i,nonCO2})} \quad \text{Equation 2}$$

Following the VM0050 methodology, the average energy consumption of baseline devices is calculated using:

$$\boldsymbol{EC_{i,y} = BC_{b,i,y} \times NCV_{b,i}} \quad \text{Equation 3}$$

Where:

$BC_{b,i,y}$	=	Fuel used per baseline device type i during year y (tonnes)
$NCV_{b,i}$	=	Net calorific value of baseline fuel for baseline device type i (TJ/tonne)

Calculations for Instance 1 at y = 0 are provided in section 4.1 of the PDD. From the PDD, $EC_{i,0}$ for Instance 1 is calculated as 0.0753 TJ. As $BC_{b,i,y}$ is fixed for 5 years, the value of $EC_{i,y}$ for Instance 1 remains unchanged and therefore $EC_{i,1} = 0.0753\ TJ$.

For Instance 2, the following parameters are used:

Parameter	Value	Source
$BC_{b,i,0}$	6.04 tonnes	Baseline KPT– Instance 2
$NCV_{b,i}$	0.0156 TJ/tonne	Methodology

$$EC_{i,0} = 6.04 \times 0.0156$$

$$= 0.0942 \text{ TJ}$$

$EC_{i,y}$ for both Instances must then be crosschecked to identify if there is any stove stacking. This is done by comparing the quantity of baseline energy consumption to energy used in the project scenario via a back-calculation with stove thermal efficiencies. If the results indicate a baseline consumption that is higher than that indicated through the back-calculation, the latter is applied as a conservative cap. This is calculated using the following equation:

$$\boldsymbol{EC_{est,y} = EC_{p,y} \times \frac{\eta_{new,avg,y}}{\eta_{old,avg}}} \quad \text{Equation 4}$$

Where:

$EC_{est,y}$	=	Back-calculated energy consumption of the potential mix of devices and fuels in the baseline in year y (TJ)
$EC_{p,y}$	=	Energy used in project scenario by project devices during year y (TJ)
$\eta_{new,avg,y}$	=	Weighted average efficiency of project devices in year y (fraction)
$\eta_{old,avg}$	=	Weighted average efficiency of baseline devices that are replaced by project devices (fraction)

First $EC_{p,y}$ is calculated using:

$$\boldsymbol{EC_{p,y} = \sum_{j,k} BC_{p,j,k,y} \times NCV_{p,j}} \quad \text{Equation 5}$$

Where:

$BC_{p,j,k,y}$	=	Average quantity of fuel used by project device type j from batch k during year y (tonnes or m³)
$NCV_{p,j}$	=	Net calorific value of project fuel used in project device type j (TJ/tonne or TJ/m³)

For this monitoring period, $EC_{est,y}$ is calculated for Instance 1 at $y = 1$ and Instance 2 at $y = 0$. We use the following parameters and apply Equation 4 to determine if a cap is applied:

Parameter	Value	Source
$BC_{p,j,1,1}$	1.66 tonnes	Project KPT– Instance 1
$BC_{p,j,2,0}$	1.82 tonnes	Project KPT– Instance 2
$NCV_{p,j}$	0.0156 TJ/tonne	Methodology
$\eta_{new,avg,0}$	36.01%	Water Boiling Test (see PDD Appendix 2)
$\eta_{new,avg,1}$	34.73%	Water Boiling Test (see Appendix 2)
$\eta_{old,avg}$	15%	Methodology

For Instance 1:

$$EC_{p,1} = 1.66 \times 0.0156$$

$$= 0.0259\ TJ$$

$$EC_{est,1} = 0.0259 \times \frac{0.3473}{0.15}$$

$$= 0.0599\ TJ$$

As $0.0599\ (EC_{est,1}) < 0.0753\ (EC_{i,1})$, a cap is applied and $EC_{i,1}$ for Instance 1 is taken as 0.0599 TJ.

For Instance 2:

$$EC_{p,0} = 1.82 \times 0.0156$$

$$= 0.0283\ TJ$$

$$EC_{est,0} = 0.0283 \times \frac{0.3601}{0.15}$$

$$= 0.0680\ TJ$$

From above, $EC_{i,0}$ for Instance 2 is calculated as 0.0942 TJ.

As $0.0680\ (EC_{est,0}) < 0.0942\ (EC_{i,y})$, $EC_{i,0}$ for Instance 2 is taken as 0.0680 TJ.

To then calculate BE_y in Equation 2, $EC_{i,y}$ here is used along with the following parameter values for the second monitoring period:

Parameter	Value	Source
$BC_{b,1,y}$	4.83 tonnes	Baseline KPT Instance 1
$BC_{b,2,y}$	6.04 tonnes	Baseline KPT Instance 2
$EF_{b,i,CO2}$	112 tCO_2/TJ	Methodology
$EF_{b,i,nonCO2}$	9.46 tCO_2/TJ	Methodology
$NCV_{b,i}$	0.0156 TJ/tonne	Methodology
$f_{NRB,1}$	0.510	Methodology Tool (see PDD section 4.4)
$f_{NRB,2}$	0.556	Methodology Tool (see section 5.4)

The calculation of BE_y for this monitoring period is presented in the sheet “ICS-01 Emission Reductions M2” and is used to determine ER_y as described in section 5.4 of this report.

5.2 Project Emissions

To calculate the project emissions, PE_y, for the second monitoring period, the approach described in section 4.2 of the PDD is followed using the following input parameters:

Parameter	Value	Source
$EF_{p,i,CO2}$	112 tCO_2/TJ	Methodology
$EF_{p,i,nonCO2}$	9.46 tCO_2/TJ	Methodology
$NCV_{p,j}$	0.0156 TJ/tonne	Methodology
$BC_{p,j,1,y}$	1.66 tonnes	Project KPT Instance 1
$BC_{p,j,2,y}$	1.82 tonnes	Project KPT Instance 2

$f_{NRB,1}$	0.510	Methodology Tool (see PDD section 4.4)
$f_{NRB,2}$	0.556	Methodology Tool (see section 5.4)

The calculation of PE_y for this monitoring period is presented in the sheet "ICS-01 Emission Reductions M2" and is used to determine ER_y as described in section 5.4 of this report.

5.3 Leakage Emissions

The following section has not changed during the monitoring period. Please see section 4.3 of the PDD for details.

5.4 GHG Emission Reductions and Carbon Dioxide Removals

To calculate the overall emissions reductions for this monitoring period, the approach described in section 4.4 of the PDD is used in combination with the following changes described below.

The fraction of non-renewable biomass for Instance 1 remains unchanged during this monitoring period. For the new project Instance 2 included in this monitoring period, f_{NRB} been calculated in line with the methodology using CDM TOOL30, including the uncertainty discount of 26%. The same process outlined in section 4.2 of the PDD is used and detailed calculations have been provided in the separate sheet "ICS-01 fNRB Calculation Instance 2"

Note that as Instance 2 is still within the Andhra Pradesh state, the only change to the inputs of the calculation is the following:

Parameter	Value	Source	Reference
HW	$6.04\ \text{tonnes/hh/year}$	Baseline Survey – Instance 2	Average pre-project fuel usage

Following the steps outlined in section 4.2 of the PDD, the fNRB for Instance 2 is calculated as:

$$f_{NRB} = 0.556$$

Together with this fNRB and the outputs from the previous sections 5.1 – 5.3, the approach described in section 4.4 of the PDD is applied to determine the emission reductions. For the second monitoring period (24th Aug 2023 – 12th Sep 2024), the total emissions reductions are calculated as $172{,}045\ tCO_2$e.

The following table summarises the baseline, project, leakage and emissions reductions over the second monitoring period, separated by vintage. Detailed calculations are provided in the separate emission reductions calculation sheet:

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Reduction VCUs (tCO_2e)	Total VCUs (tCO_2e)
24-Aug-2023 to 31-Dec-2023	66,347	27,637	1,935	36,775	36,775
01-Jan-2024 to 12-Sep-2024	245,643	103,253	7,120	135,270	135,270
Total	311,990	130,890	9,055	172,045	172,045

The table below then compares the achieved emissions reductions (stated in the table above) to that estimated from ex-ante calculations. Differences in reductions are explained.

Vintage period	Ex-ante estimated reductions/ removals	Achieved reductions/ removals	Percent difference	Explanation for the difference
24-Aug-2023 to 31-Dec-2023	67,034	36,775	−45.1%	The percentage difference in emissions reductions can mainly be attributed to a conservative cap applied to the baseline energy consumption parameter $EC_{i,y}$ under the VM0050 methodology. For ex-ante purposes, $EC_{i,0}$ was taken as 0.0753 TJ which was estimated using the efficiencies of the baseline and project stoves. However, for ex-post a cap of 0.0621 TJ is used based on the results of Kitchen Performance Test (KPT) results, leading to fewer emissions reductions.
01-Jan-2024 to 12-Sep-2024	283,923	135,270	−52.4%	During 2024, the difference in emissions reductions was primarily due to extreme weather from June to September, resulting in the project area being inaccessible for large parts of these months. The bad weather (over the monsoon period) was particularly unprecedented and represents the negative effects of climate change. More specifically, during the period the daily distribution rate dropped

				from approximately 900 per day to as low as 5 per day, which had a significant impact on the planned rollout. Further, there was India's general elections held between April and June 2024, which prevented any distribution for a period over this time. Since the first monitoring period, the team have gained more experience in surveying and this has resulted in a drastically reduced number of errors during data collection. An additional reason for this difference is that up until the end of the monitoring period, only CRS-23 stoves have been distributed. In the ex-ante model, it was assumed that distribution of the more efficient STAR-25 would have begun in project year 2.
Total	$350{,}957$	$172{,}045$	-51.0%	

APPENDIX 1

The schedule followed, including the event details for stakeholder meetings conducted for Instance 2 in the Rampachodavaram division of Andhra Pradesh, is outlined below:

Mandal	Village	Meeting Venue	Attendees	Date	Special Guests
Addateegala	Rayapalli	Near Ramalayam	252	8-Nov-2023	Chedala Viswanadham (Raypalli Panchayat President) Aguri Veeravenkata Satyanrayana Reddy (Veeravaram President) Madakam Satyanarayana (MPTC Member)
Chinturu	Thummala	Govt School Building	311	15-Nov-2023	Kovvasi Rama Rao (President, Sarivela Village) Meka Prasad (President, Vekavaarigudem Village)
Devipatnam	Devipatnam	Manture R R Colony	746	13-Nov-2023	Turasam Srinivasu Dora (Manturu Convenor) Thokala Nagarathnam (Panchayath President) Kunjam Murali (Mandal Parisith President) Sirasam Peddabai (ZPTC Member) Tellam Shekar (Aadivasi JSC)
Gangavaram	Mohanapuram	Ramalayam Temple	255	11-Nov-2023	Kurasam Akkamma (Panchayat President) Maddeti Venkata Lakshmi (MPTC Member)

					Pallala Krishna Reddy (MPP, Marripalem Village)
Kunavaram	Sabharikot hagudem	Village Rachhaband a	281	17-Nov-2023	Kattam Lakshmi (Panchayat President) Sunnam Abhi Ram (Panchayath President) Kunja Srinu (Convenor, Sachivalayam)
Maredumilli	Maredumill i	Anganwadi Building	674	14-Nov-2023	Konda Jacob (Maredumilli Village Panchayat President) E Malleshwari (Vetukuru Village Panchayat President) K Laxmi Parvati (GM Valasa, Panchayat President)
Rajavommangi	Kirrabu	Kirrabu Colony Ramalayam	306	9-Nov-2023	Thamarapu Anantha Lakshmi (Boddagondi Ward Member) Borra Bujjamma (Kirrabu Panchayat President) Busari Satyanarayana (Kindra Panchayat President)
Rampachodav aram	Rampacho davaram	PMRC Guest House	305	28-Nov-2023	Sri. Srinivasa Rao (Assistant Project Officer, ITDA) Ch. Sridevi (Rampachodavaram Block President) Sri. Sirimalli Reddy (Tribal District Leader)

					Sri. Ramaswami (Ward Member) Ch. Swaroopini (Ward Member)
V R Puram	Adavi Venkannagudem	Main Road (Open Place)	287	16-Nov-2023	Karam Buchamma (Panchayat President) K Lakshmi (Mandal Parisith President) Punem Pradeep (ZPTC Member)
Y Ramavaram	Thotakurapalem	Grama Sachivalayam	438	10-Nov-2023	Bachala Chinnammulu (Thotakura Panchayat President) Mutyala Nookaraju (Devaramadugu Village Panchayat President) Pallala Abbai Reddy (Former President of Cheddipalem Village)
Yetapaka	Krishnavaram	Govt School Building	695	18-Nov-2023	Kursam Krishna (President, Krishnavaram Panchayat President) Chichadi Ramarao (Community Leader, Krishnavaram)

APPENDIX 2

The WBT results for 1 year old CRS-23 stoves which are used in the project areas are presented below. Tests were undertaken by a specialist external independent technical laboratory Koshish Sustainable Solutions Pvt Ltd during September 2024.

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Ref:210924/006 Date:21/09/2024

1. Water Boiling test results of Stove Model-CRS-23_1

Stove Model-CRS-23_1, a natural draft biomass cook stove was tested in the Biomass cookstove Laboratory of the Koshish Sustainable Solutions Pvt Ltd. Thestove was tested according to the water boiling test protocol WBT 4.2.3. The details of the stove and the test results are given below.

1.1 Details of Stove Model-CRS-23_1

Specification

Fuel Type	Wood
Feeding Type	Front Feeding
Combustion chamber	Ceramic
Cooking suitable for	5 to 10 persons
Draft	Natural draft
Net wet	9 Kg Approx. (Without wood stand)

QR Code: A1A0024142

1.2 Test Results

The average thermal efficiency of the Stove Model-CRS-23_1 was found to be 34.7 %. The summary of the Test results is given in the table below:

1. HIGH POWER TEST (COLD START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	21	21	22	21.3
Temp-corrected time to boil Pot # 1	min	22	22	23	22.4
Burning rate	g/min	18	18	17	17.4
Thermal efficiency	%	33.9%	33.4%	33.5%	33.6%
Specific fuel consumption	g/liter	77	77	78	77.4
Temp-corrected specific consumption	g/liter	79	82	83	81.4
Temp-corrected specific energy cons.	kJ/liter	1,232	1,269	1,290	1,264.0
Firepower	watts	4,575	4,545	4,408	4,509.3

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2. HIGH POWER TEST (HOT START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	18	19	19	18.7
Temp-corrected time to boil Pot # 1	min	19	20	20	19.7
Burning rate	g/min	18	18	19	18.4
Thermal efficiency	%	36.1%	35.8%	35.4%	35.8%
Specific fuel consumption	g/liter	66	72	77	71.6
Temp-corrected specific consumption	g/liter	68	77	83	75.8
Temp-corrected specific energy cons.	kJ/liter	1,049	1,189	1,293	1,176.8
Firepower	watts	4,589	4,685	5,004	4,759.4

IWA PERFORMANCE METRICS	units	Test 1	Test 2	Test 3	Average
High Power Thermal Efficiency	%	35.0%	34.6%	34.4%	34.7%

IWA PERFORMANCE TIERS	Tier
High Power Thermal Efficiency	2

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Ref:041024/002 Date:04/10/2024

1. Water Boiling test results of Stove Model-CRS-23_2

Stove Model-CRS-23_2, a natural draft biomass cook stove was tested in the Biomass cookstove Laboratory of the Koshish Sustainable Solutions Pvt Ltd. The stove was tested according to the water boiling test protocol WBT 4.2.3. The details of the stove and the test results are given below.

1.1 Details of Stove Model-CRS-23_2

Specification	
Fuel Type	Wood
Feeding Type	Front Feeding
Combustion chamber	Ceramic
Cooking suitable for	5 to 10 persons
Draft	Natural draft
Net wet	9 Kg Approx. (Without wood stand)

QR Code: A1A0031378

1.2 Test Results

The average thermal efficiency of the Stove Model-CRS-23_2 was found to be 34.7 %. The summary of the Test results is given in the table below:

1. HIGH POWER TEST (COLD START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	20	21	22	21.0
Temp-corrected time to boil Pot # 1	min	21	22	23	22.1
Burning rate	g/min	19	17	17	17.7
Thermal efficiency	%	33.8%	34.0%	34.3%	34.0%
Specific fuel consumption	g/liter	78	76	78	77.2
Temp-corrected specific consumption	g/liter	80	81	83	81.2
Temp-corrected specific energy cons.	kJ/liter	1,246	1,250	1,288	1,261.6
Firepower	watts	4,852	4,475	4,386	4,571.2

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2. HIGH POWER TEST (HOT START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	18	19	19	18.7
Temp-corrected time to boil Pot # 1	min	19	20	20	19.7
Burning rate	g/min	18	18	19	18.6
Thermal efficiency	%	35.2%	35.3%	35.4%	35.3%
Specific fuel consumption	g/liter	67	73	77	72.3
Temp-corrected specific consumption	g/liter	69	78	83	76.5
Temp-corrected specific energy cons.	kJ/liter	1,065	1,214	1,285	1,187.9
Firepower	watts	4,670	4,775	4,979	4,807.9

IWA PERFORMANCE METRICS	units	Test 1	Test 2	Test 3	Average
High Power Thermal Efficiency	%	34.5%	34.7%	34.8%	34.7%

IWA PERFORMANCE TIERS	Tier
High Power Thermal Efficiency	2

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Ref:041024/003 Date:04/10/2024

1. Water Boiling test results of Stove Model-CRS-23_3

Stove Model-CRS-23_3, a natural draft biomass cook stove was tested in the Biomass cookstove Laboratory of the Koshish Sustainable Solutions Pvt Ltd. Thestove was tested according to the water boiling test protocol WBT 4.2.3. The details of the stove and the test results are given below.

1.1 Details of Stove Model-CRS-23_3

Specification	
Fuel Type	Wood
Feeding Type	Front Feeding
Combustion chamber	Ceramic
Cooking suitable for	5 to 10 persons
Draft	Natural draft
Net wet	9 Kg Approx. (Without wood stand)

QR Code: A1A0008699

1.2 Test Results

The average thermal efficiency of the Stove Model-CRS-23_3 was found to be 34.6 %. The summary of the Test results is given in the table below:

1. HIGH POWER TEST (COLD START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	22	21	20	21.0
Temp-corrected time to boil Pot # 1	min	23	22	21	22.1
Burning rate	g/min	17	17	19	17.8
Thermal efficiency	%	33.2%	34.0%	34.3%	33.8%
Specific fuel consumption	g/liter	79	76	78	77.7
Temp-corrected specific consumption	g/liter	82	81	83	81.7
Temp-corrected specific energy cons.	kJ/liter	1,269	1,250	1,288	1,269.0
Firepower	watts	4,489	4,475	4,825	4,596.2

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2. HIGH POWER TEST (HOT START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	18	19	19	18.7
Temp-corrected time to boil Pot # 1	min	19	20	20	19.7
Burning rate	g/min	18	18	19	18.5
Thermal efficiency	%	35.5%	35.3%	35.4%	35.4%
Specific fuel consumption	g/liter	66	73	77	72.0
Temp-corrected specific consumption	g/liter	68	78	83	76.2
Temp-corrected specific energy cons.	kJ/liter	1,053	1,214	1,285	1,183.7
Firepower	watts	4,616	4,775	4,979	4,789.8

IWA PERFORMANCE METRICS	units	Test 1	Test 2	Test 3	Average
High Power Thermal Efficiency	%	34.4%	34.7%	34.8%	34.6%

IWA PERFORMANCE TIERS	Tier
High Power Thermal Efficiency	2

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Ref:041024/004 **Date:04/10/2024**

1. Water Boiling test results of Stove Model-CRS-23_4

Stove Model-CRS-23_4, a natural draft biomass cook stove was tested in the Biomass cookstove Laboratory of the Koshish Sustainable Solutions Pvt Ltd. The stove was tested according to the water boiling test protocol WBT 4.2.3. The details of the stove and the test results are given below.

1.1 Details of Stove Model-CRS-23_4

Specification	
Fuel Type	Wood
Feeding Type	Front Feeding
Combustion chamber	Ceramic
Cooking suitable for	5 to 10 persons
Draft	Natural draft
Net wet	9 Kg Approx. (Without wood stand)

QR Code: A1A0007441

1.2 Test Results

The average thermal efficiency of the Stove Model-CRS-23_4 was found to be 34.9 %. The summary of the Test results is given in the table below:

1. HIGH POWER TEST (COLD START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	21	21	21	21.0
Temp-corrected time to boil Pot # 1	min	22	22	22	22.1
Burning rate	g/min	18	17	18	17.7
Thermal efficiency	%	33.6%	33.8%	33.5%	33.6%
Specific fuel consumption	g/liter	78	75	80	77.4
Temp-corrected specific consumption	g/liter	80	80	85	81.5
Temp-corrected specific energy cons.	kJ/liter	1,242	1,238	1,317	1,265.4
Firepower	watts	4,610	4,440	4,700	4,583.2

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2. HIGH POWER TEST (HOT START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	17	19	19	18.3
Temp-corrected time to boil Pot # 1	min	18	20	20	19.4
Burning rate	g/min	18	18	19	18.4
Thermal efficiency	%	36.4%	36.3%	36.0%	36.2%
Specific fuel consumption	g/liter	64	72	76	70.5
Temp-corrected specific consumption	g/liter	66	77	81	74.6
Temp-corrected specific energy cons.	kJ/liter	1,017	1,196	1,262	1,158.4
Firepower	watts	4,730	4,698	4,889	4,772.1

IWA PERFORMANCE METRICS	units	Test 1	Test 2	Test 3	Average
High Power Thermal Efficiency	%	35.0%	35.0%	34.8%	34.9%

IWA PERFORMANCE TIERS	Tier
High Power Thermal Efficiency	2

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1. Water Boiling test results of Stove Model-CRS-23_5

Stove Model-CRS-23_5, a natural draft biomass cook stove was tested in the Biomass cookstove Laboratory of the Koshish Sustainable Solutions Pvt Ltd. Thestove was tested according to the water boiling test protocol WBT 4.2.3. The details of the stove and the test results are given below.

1.1 Details of Stove Model-CRS-23_5

Specification	
Fuel Type	Wood
Feeding Type	Front Feeding
Combustion chamber	Ceramic
Cooking suitable for	5 to 10 persons
Draft	Natural draft
Net wet	9 Kg Approx. (Without wood stand)

QR Code: A1A0019697

1.2 Test Results

The average thermal efficiency of the Stove Model-CRS-23_5 was found to be 35.0 %. The summary of the Test results is given in the table below:

1. HIGH POWER TEST (COLD START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	21	21	22	21.3
Temp-corrected time to boil Pot # 1	min	22	22	23	22.4
Burning rate	g/min	18	17	17	17.3
Thermal efficiency	%	33.6%	33.9%	33.8%	33.8%
Specific fuel consumption	g/liter	78	74	79	77.0
Temp-corrected specific consumption	g/liter	80	79	84	81.1
Temp-corrected specific energy cons.	kJ/liter	1,245	1,231	1,300	1,258.8
Firepower	watts	4,621	4,417	4,431	4,489.6

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2. HIGH POWER TEST (HOT START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	18	18	19	18.3
Temp-corrected time to boil Pot # 1	min	19	19	20	19.4
Burning rate	g/min	18	19	19	18.4
Thermal efficiency	%	36.0%	36.4%	36.0%	36.1%
Specific fuel consumption	g/liter	65	71	76	70.4
Temp-corrected specific consumption	g/liter	67	76	81	74.5
Temp-corrected specific energy cons.	kJ/liter	1,033	1,178	1,262	1,157.6
Firepower	watts	4,535	4,891	4,889	4,771.5

IWA PERFORMANCE METRICS	units	Test 1	Test 2	Test 3	Average
High Power Thermal Efficiency	%	34.8%	35.2%	34.9%	35.0%

IWA PERFORMANCE TIERS	Tier
High Power Thermal Efficiency	2

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Ref:041024/006 Date:04/10/2024

1. Water Boiling test results of Stove Model-CRS-23_6

Stove Model-CRS-23_6, a natural draft biomass cook stove was tested in the Biomass cookstove Laboratory of the Koshish Sustainable Solutions Pvt Ltd. The stove was tested according to the water boiling test protocol WBT 4.2.3. The details of the stove and the test results are given below.

1.1 Details of Stove Model-CRS-23_6

Specification	
Fuel Type	Wood
Feeding Type	Front Feeding
Combustion chamber	Ceramic
Cooking suitable for	5 to 10 persons
Draft	Natural draft
Net wet	9 Kg Approx. (Without wood stand)

QR Code: A1A0010594

1.2 Test Results

The average thermal efficiency of the Stove Model-CRS-23_6 was found to be 34.5 %. The summary of the Test results is given in the table below:

1. HIGH POWER TEST (COLD START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	21	21	22	21.3
Temp-corrected time to boil Pot # 1	min	22	22	23	22.4
Burning rate	g/min	18	17	17	17.6
Thermal efficiency	%	33.4%	33.5%	33.1%	33.3%
Specific fuel consumption	g/liter	79	76	80	78.3
Temp-corrected specific consumption	g/liter	81	81	85	82.4
Temp-corrected specific energy cons.	kJ/liter	1,258	1,255	1,326	1,279.5
Firepower	watts	4,668	4,498	4,519	4,561.7

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2. HIGH POWER TEST (HOT START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	18	19	19	18.7
Temp-corrected time to boil Pot # 1	min	19	20	20	19.7
Burning rate	g/min	18	18	19	18.3
Thermal efficiency	%	35.6%	35.9%	35.7%	35.7%
Specific fuel consumption	g/liter	66	72	75	71.0
Temp-corrected specific consumption	g/liter	68	77	81	75.1
Temp-corrected specific energy cons.	kJ/liter	1,053	1,190	1,255	1,166.1
Firepower	watts	4,616	4,685	4,876	4,725.7

IWA PERFORMANCE METRICS	units	Test 1	Test 2	Test 3	Average
High Power Thermal Efficiency	%	34.5%	34.7%	34.4%	34.5%

IWA PERFORMANCE TIERS	Tier
High Power Thermal Efficiency	2

Tested and Report prepared by

Amit Ranjan Verma

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Date: 2024.10.08 17:25:46 +05'30'

Dr. Amit Ranjan Verma (PhD-IIT Delhi)

Adviser and Energy Expert

(Expertise in Biomass Cookstove Testing by using National and International Protocols such as WBT_BIS, WBT 4.2.3, CCT, KPT, UCT)

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